ESG Revenue Screener

Product Proposal: Configurable, Evidence-Based Screening of Company Revenue Activities

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1. Executive Summary

The ESG Revenue Screener is a configurable tool designed to help users assess and compare UK companies against their own environmental, social and governance (ESG) investment preferences.

The proposed product addresses a practical challenge: users may want to screen companies against specific activities, such as renewable energy, healthcare, fossil fuels or tobacco, but need to understand how much revenue is associated with those activities and what evidence supports the assessment.

Rather than assigning companies a universal ESG score, the system will allow users to select predefined positive and exclusionary activities, configure individual revenue thresholds and apply their screening profiles to a dataset of 20 UK companies.

The solution will automate the retrieval and analysis of company accounts through the Companies House API, using document extraction and AI to identify relevant activities and financial information. Deterministic screening logic will then calculate revenue exposure and classify each selected criterion as Green, Red or Amber.

A Streamlit dashboard, supported by a PostgreSQL database, will present company comparisons, individual screening outcomes and traceable evidence from original filings. Users will also be able to save and reuse multiple screening profiles.

The initial minimum viable product (MVP) will prioritise transparency, reliable extraction, configurable screening and evidence traceability. A conversational AI assistant for creating screening profiles and investigating company results will be considered as a future enhancement.

2. Problem Statement and Rationale

ESG screening can help investors align investment decisions with particular sustainability preferences, values and risk considerations. The Principles for Responsible Investment (PRI,

2023) identifies positive and exclusionary screening as established responsible-investment approaches. Positive screening identifies investments associated with desirable ESG characteristics, while exclusionary screening applies rules to activities or characteristics an investor wishes to avoid.

However, ESG assessments are not necessarily consistent across providers. Berg, Kölbel and Rigobon (2022) examined six major ESG rating agencies and found that differences in measurement accounted for 56% of rating divergence, differences in scope for 38%, and differences in weighting for 6%. This highlights the importance of understanding how ESG information is selected and measured.

The proposed screener addresses this challenge by allowing users to define their own activity-based screening criteria and revenue thresholds. It also makes the underlying evidence visible, rather than presenting an unexplained score.

There is, however, an important data limitation. Companies House filings do not consistently provide detailed revenue breakdowns. Small companies and micro-entities may file accounts without a publicly available profit and loss account (Companies House, 2026). Furthermore, revenue disclosure requirements and levels of disaggregation depend on the applicable reporting framework and circumstances. For example, IFRS 15 requires revenue disaggregation into categories that reflect relevant economic factors, but does not prescribe a universal ESG activity breakdown (IFRS Foundation, 2024).

Consequently, the product must distinguish between a confirmed absence of exposure and an absence of sufficient evidence.

3. Proposed Solution and Product Objectives

The ESG Revenue Screener will provide five main capabilities:

1. **Configurable screening:** Users select predefined ESG activities and set individual revenue thresholds.
2. **Automated data extraction:** Company accounts are retrieved and analysed to identify relevant activities, financial figures and supporting evidence.
3. **Consistent classification:** Deterministic rules apply the selected thresholds and produce Green, Red or Amber results.
4. **Company comparison:** Users compare multiple companies through a dashboard and investigate individual results.
5. **Evidence traceability:** Screening outcomes link back to the original filing, relevant passage and financial calculation.

The product is intended to support research and investment screening, not to provide financial advice or establish whether a company is universally sustainable, ethical or responsible.

4. ESG Screening Framework

The MVP will use a predefined activity library covering Environmental, Social and limited Governance indicators.

4.1 Environmental (E)

Activity	Screening type	Initial scope
Renewable energy	Positive	Renewable energy generation, including wind and solar
Recycling and waste management	Positive	Waste collection, recycling, processing and treatment
Fossil fuels	Exclusionary	Coal, oil and gas-related activities
Deforestation	Exclusionary	Activities directly associated with deforestation

Oil and gas extraction will be treated as a subcategory of fossil fuels to avoid duplicating overlapping criteria.

4.2 Social (S)

Activity	Screening type	Initial scope
Healthcare	Positive	Healthcare services and treatment
Medical research	Positive	Medical and pharmaceutical research
Affordable housing	Positive	Development and provision of affordable housing
Education	Positive	Education and training services
Tobacco	Exclusionary	Tobacco production and related products
Gambling	Exclusionary	Gambling operations and services
Weapons	Exclusionary	Weapons manufacturing and related activities
Alcohol	Exclusionary	Alcohol production and sales

Activity definitions will specify whether manufacturing, distribution, retail and supporting services are included. Weapons, in particular, will require a clear definition of whether the exclusion applies to all weapons or only specified categories.

4.3 Governance (G)

Governance will have limited coverage in the MVP through disclosure-based indicators, including:

- Board structure and oversight disclosures.
- Audit arrangements and financial reporting disclosures.
- Risk-management and internal-control disclosures.
- Corporate governance statements.

These indicators will be presented separately from revenue-based screening. Their presence does not automatically demonstrate effective governance, and company accounts alone cannot establish comprehensive governance quality or identify every instance of misconduct.

Broader governance and controversy screening will require additional data sources and is therefore a future development area.

5. Configurable Screening Methodology

5.1 Screening profiles and thresholds

Users will create and save multiple screening profiles. Each profile will contain selected activities and individually configured thresholds.

Positive activities will use a minimum revenue threshold, while exclusionary activities will use a maximum permitted revenue threshold.

For example:

Activity	Type	Illustrative threshold
Renewable energy	Minimum	15%
Healthcare	Minimum	10%
Medical research	Minimum	5%
Fossil fuels	Maximum	5%
Tobacco	Maximum	0%

Gambling	Maximum	2%

These values are illustrative only. Users will choose their own thresholds when creating a profile; the MVP will not impose universal default percentages.

5.2 Revenue exposure calculation

Where the accounts provide sufficient information, activity exposure will be calculated as:

Activity revenue exposure (%) = (Revenue attributable to activity / Total company revenue) × 100

The numerator and denominator must be comparable in reporting period, currency, reporting entity and accounting basis. The system will record the source of both figures and validate financial consistency before applying a threshold.

Where revenue cannot be separately identified or reliably attributed to an activity, the system will not invent an estimate.

5.3 Direct and indirect involvement

The system will distinguish between:

- **Direct involvement:** The company itself performs the relevant activity, such as extracting oil or manufacturing tobacco.
- **Indirect involvement:** The company supplies, distributes or otherwise supports an organisation involved in the activity.

Both types will be identified where supporting evidence is available. However, indirect involvement will only contribute to a calculated revenue exposure where the relevant revenue is separately disclosed and measurable. Otherwise, the system will record the evidence and explain why the percentage-based assessment is inconclusive.

5.4 Screening outcomes

Status	Positive screening	Exclusionary screening
Green	Revenue meets or exceeds the minimum threshold	Revenue is within the maximum permitted threshold
Red	Sufficient evidence demonstrates revenue is below the minimum	Revenue exceeds the maximum permitted threshold
Amber	Evidence is insufficient or ambiguous	Evidence is insufficient or ambiguous

Amber outcomes will include an explanation, such as missing revenue detail, ambiguous activity descriptions, uncertain extraction or failed financial reconciliation.

5.5 Overall company outcome

An overall profile-specific outcome will be calculated from the selected revenue criteria:

- **Green:** All selected criteria are Green.
- **Red:** At least one selected criterion is Red.
- **Amber:** No criteria are Red, but at least one is Amber.

Governance disclosure indicators will remain separate from this calculation unless a distinct and justified aggregation methodology is developed.

The overall outcome is not a universal ESG rating. It only describes how a company performs against the particular profile and thresholds selected by the user.

6. User Journey and Dashboard

The Streamlit dashboard will provide three main views.

1. Screening overview

Users can view their active profile, number of companies screened, overall Green/Red/Amber counts, latest screening date and available filters.

2. Company comparison matrix

Companies will appear as rows and selected ESG activities as columns. Each cell will display its criterion-level classification. Users can compare all 20 companies or select an individual subset.

3. Individual company detail

Selecting a company or result will display:

- Company name and Companies House identifier.
- Filing name, filing date and financial reporting period.
- Relevant ESG activity and direct/indirect involvement.
- Activity revenue and total revenue, where disclosed.
- Calculated revenue exposure and applied threshold.
- Green, Red or Amber result and explanation.
- Supporting passage, PDF page reference and original filing link.
- Reason for any inconclusive result.

This structure makes the comparison matrix useful for identifying patterns while allowing users to investigate the evidence behind individual classifications.

7. Technical Architecture

The proposed architecture consists of five main components.

Component	Responsibility
Companies House API	Retrieve company information, filing history and available filing documents
n8n	Orchestrate API requests, document processing and screening workflows
PDF extraction and AI	Identify business activities, revenue information and supporting passages
Python/JavaScript	Validate financial information, calculate revenue exposure and apply classification rules
PostgreSQL and Streamlit	Store structured information and provide the user-facing dashboard

Companies House provides API access to public company information and filing history. Its Document API provides filing document metadata, including available content types and page counts; PDF availability varies by filing .

The workflow will:

1. Retrieve a company's filing history.
2. Identify the latest suitable accounts filing.
3. Download and store the available PDF.
4. Extract relevant financial figures, activities and evidence.
5. Validate extracted information.
6. Apply the selected profile using deterministic screening logic.
7. Store results and display them through Streamlit.

AI will support extraction and interpretation, but will not independently determine final screening outcomes. Threshold comparisons and Green/Red/Amber decisions will be implemented in deterministic code.

The dashboard will use Streamlit's multipage functionality to separate screening overview, comparison and company detail views (Streamlit, n.d.). PostgreSQL will provide relational data storage, with primary keys, foreign keys and appropriate constraints supporting data integrity (PostgreSQL Global Development Group, 2026).

8. Database, Evidence and Auditability

The database will separate original filings, extracted information, user profiles and screening results.

Proposed table	Purpose
`companies`	Company details and Companies House identifiers
`filings`	Filing metadata, document references and reporting periods
`revenue_streams`	Extracted activities and associated financial values
`screening_profiles`	Saved user screening profiles
`criteria_versions`	Historical profile configurations
`criteria`	Predefined ESG activities and classifications
`screening_runs`	Individual screening executions
`classifications`	Criterion-level results
`evidence`	Supporting passages, page references and source links
`validation_results`	Extraction and financial validation checks
`audit_log`	Record of significant actions and changes
`governance_indicators`	Separate governance disclosure findings

Original documents and extracted information will be stored separately from screening results. This allows users to apply different profiles to the same extracted company data without repeatedly processing unchanged documents.

Every screening result will retain the relevant filing reference, reporting period, profile version, criteria thresholds and calculation inputs. The audit log will record significant system actions, but will not be described as tamper-proof unless additional security controls are implemented.

9. Filing Updates and Data Quality

Whenever a new screening is initiated, the system will check Companies House for newer suitable accounts filings.

If a newer filing is found, the system will retrieve and process it. If the existing filing remains the latest suitable document, stored extracted information can be reused. Changes to screening thresholds should not require document reprocessing.

The financial reporting period must be stored separately from the filing date. This prevents users from confusing when accounts were submitted with the period their financial information covers.

The main data-quality risks are:

- Missing or limited financial disclosures.
- Revenue figures not disaggregated by relevant ESG activity.
- Ambiguous business activity descriptions.
- AI extraction errors.
- Overlapping activity definitions and double-counting.
- Indirect involvement that cannot be quantified.
- Inconsistent financial periods, currencies or reporting entities.

These risks will be addressed through validation checks, clearly defined activity categories, source traceability and Amber outcomes. Missing information will never automatically be interpreted as zero exposure.

10. MVP Scope and Stretch Goals

MVP deliverables

The initial product will include:

- Predefined Environmental and Social revenue activities and limited Governance disclosure indicators.
- Positive and exclusionary screening.
- Individually configurable revenue thresholds.
- Saved and reusable screening profiles.
- An initial dataset of 20 UK companies.
- Companies House filing history integration and PDF retrieval.
- Automated activity and revenue extraction.
- Direct and indirect involvement identification, with appropriate quantification limitations.
- Financial validation and deterministic classification.
- Green, Red and Amber criterion-level outcomes.
- Overall profile-specific company outcomes.
- Streamlit screening overview, comparison matrix and company detail pages.
- PostgreSQL storage, evidence traceability and basic audit logging.

- Profile and criteria version history.
- Checks for newer suitable filings whenever a screening is initiated.

Stretch goals

Priority	Feature	Purpose
1	Screening profile chatbot	Convert natural-language preferences into structured profiles, with user confirmation
2	Company research chatbot	Answer questions using stored screening results and evidence
3	Expanded governance screening	Assess additional governance indicators using external sources
4	XHTML/iXBRL processing	Improve extraction from structured accounts
5	Larger company dataset	Expand beyond the initial 20 companies
6	Year-on-year comparisons	Track changes in company revenue exposure
7	Advanced indirect involvement	Improve identification and quantification where evidence permits
8	Company ranking and weighting	Introduce comparisons beyond criterion-level outcomes using an explicit methodology
9	Downloadable reports	Provide exportable company and portfolio screening reports

The chatbot is deliberately excluded from the MVP. Its first proposed use is assisting with profile creation rather than making autonomous screening decisions.

11. Evaluation and Success Criteria

The MVP will be evaluated using a manually verified sample of company accounts. This will allow system outputs to be compared against the original documents.

Evaluation area	Proposed test
Revenue extraction	Compare extracted figures against manually checked accounts

Activity identification	Assess whether relevant activities are correctly identified
Classification accuracy	Compare Green, Red and Amber outcomes against expected test cases
Financial calculations	Verify revenue exposure percentages independently
Threshold logic	Test minimum, maximum and boundary conditions
Evidence traceability	Confirm each result links to identifiable source evidence or an explicit Amber reason
Profile configurability	Confirm different profiles and thresholds produce expected results
Filing updates	Test detection of newer suitable filings and reuse of unchanged data
Auditability	Confirm results can be traced to the filing and profile version used
Dashboard usability	Test whether users can create profiles, screen companies and investigate results

Performance targets should be agreed after establishing the test dataset. No extraction or classification accuracy should be assumed before validation.

12. Implementation Plan

Phase	Activities	Deliverable
1. Foundations	Finalise activity definitions, company dataset and database schema	Screening specification and data model
2. Data ingestion	Connect Companies House API, retrieve filing history and download PDFs	Working filing ingestion
3. Extraction	Extract activities, revenue figures and supporting evidence	Structured company information
4. Screening engine	Implement financial validation, threshold calculations and classification rules	Tested screening logic
5. Dashboard	Build profile management, company comparison and evidence views	Functional Streamlit application

6. Testing	Validate results, filing updates, auditability and usability	Evaluation findings and demonstration

Development should prioritise a complete end-to-end workflow for a small number of companies before expanding to all 20.

13. Expected Value and Conclusion

The ESG Revenue Screener will provide a practical way to apply configurable, activity-based screening rules to publicly available company information.

Its value lies in combining user-defined investment preferences with automated data extraction, consistent calculations and transparent evidence. Rather than hiding methodological choices inside a single ESG score, the product will make the selected criteria, thresholds and supporting information visible.

The project also demonstrates a controlled use of AI: AI assists with interpreting unstructured accounts, while deterministic logic handles financial calculations and screening decisions.

The main limitation is the availability and quality of public revenue disclosures. The system will therefore communicate uncertainty explicitly rather than making unsupported assumptions.

By prioritising configurability, traceability and defensible decision logic, the MVP will provide a focused foundation that can later be extended with conversational assistance, broader governance data and more sophisticated company comparisons.

References

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